Formation of a Binary Star

Since it is impossible to observe the early ages of binary stars, scientists have yet to explore how they are formed. One of the explanations for the emergence of binary stars suggests that a massive star seizes another star that is passing by.

However, this is considered to be a rare event. The most commonly suggested theory is that binaries are shaped during a star's birth. Given that stars form massive gas clouds consisting of interstellar gas, gravity starts to pull the gas inwards and collapses it to form a new star.

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Binary Star

Understanding Binary Stars

Star A and Star B

From the physics perspective, binary stars are two stars orbiting around their common center of mass. The brightest star among the two is called a primary star, or star A, while the dimmer one is the secondary star, or star B (also called "the companion").

Close and Wide Binaries

The stars are often categorized based on their orbit. There are close and wide binaries where the first case allows a particular impact of one star to the other by transferring masses between them. At the same time, the latter presumes no or minimal impact on one another. Some binaries are so close that they can touch each other. Astronomers call these "contact binaries".

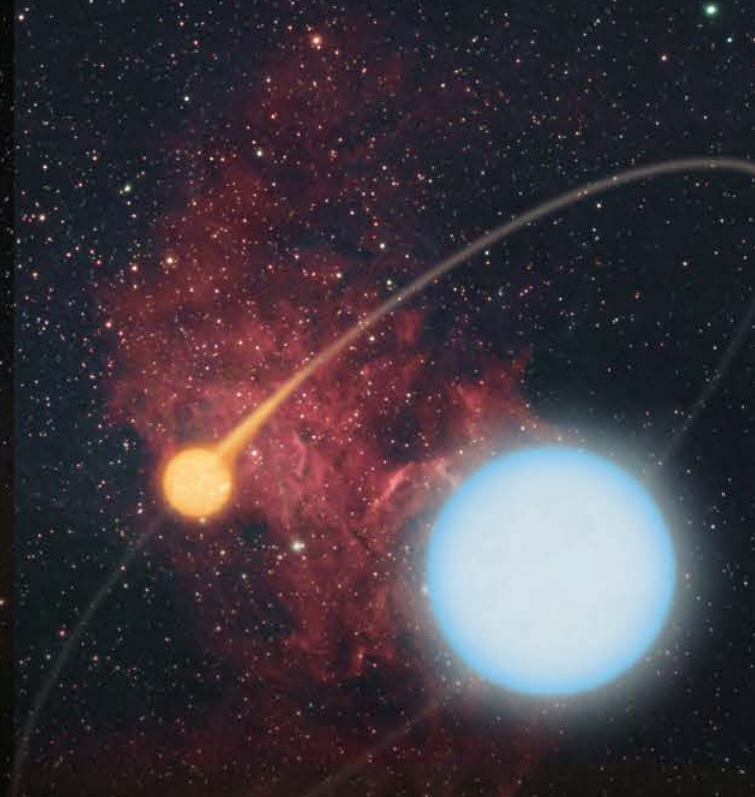
Classification by Observation Method

Visual. Observing the separation between stars is possible using a telescope or a pair of binoculars.

Spectroscopic. This type dictates that the visual separation is minimal. Scientists can separate them only by wavelengths of the light these stars project.

Eclipsing. The distance between the stars can be significant enough to see the separation. Still, the angle is such that one star is in front of the other when viewing from the Earth.

Astrometric. It might seem that one of two stars orbits around empty space. Still, the companion is either too dim or cannot be seen because of the glare of the primary star.



Other Variations

The Shape of the Orbit

Binary stars can vary in many other ways. For instance, binaries can differ depending on the shape of the orbits they revolve around. Some binaries might travel around the shape of an ellipse, while others might do circular movements.

Age Difference

Binary pairs can consist of stars with a considerable age difference. In this case, as one of two stars ages, it can lose some of its material due to stellar winds. In such an event, the other star can obtain that material through gravity.